

Lecture 7

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Review of gradient, Jacobian, and derivative definitions

Recall our preferred definitions of $\nabla f(\mathbf{x})$, $\nabla f(X)$, $Jf(\mathbf{x})$:

$f : \mathbb{R}^n \rightarrow \mathbb{R}$, whatever vector you can put in the box below

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \square^\top \mathbf{h}|}{\|\mathbf{h}\|} = 0$$

so that the limit holds is the gradient of f at $\mathbf{x}$;

$f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$, whatever matrix you can put in the box below

$$\lim_{H \rightarrow 0} \frac{|f(X + H) - f(X) - \text{tr}(\square^\top H)|}{\|H\|} = 0$$

so that the limit holds is the gradient of f at X ;

Put in terms of the derivative:

$f : \mathbb{R}^n \rightarrow \mathbb{R}$, whatever vector you can put in the box below

$$[Df(\mathbf{x})](\mathbf{h}) = \square^\top \mathbf{h}$$

is the gradient of f at $\mathbf{x}$;

$f : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}$, whatever matrix you can put in the box below

$$[Df(X)](H) = \text{tr}(\square^\top H)$$

is the gradient of f at X

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, whatever matrix you can put in the box below

$$\lim_{\mathbf{h} \rightarrow 0} \frac{\|f(\mathbf{x} + \mathbf{h}) - f(\mathbf{x}) - \square \mathbf{h}\|}{\|\mathbf{h}\|} = 0$$

so that the limit holds is the Jacobian of f at $\mathbf{x}$.

$f : \mathbb{R}^n \rightarrow \mathbb{R}^m$, whatever matrix you can put in the box below

$$[Df(\mathbf{x})](\mathbf{h}) = \square \mathbf{h}$$

is the Jacobian of f at $\mathbf{x}$.

Examples of applying the derivative rules

Example: quadratic function

Let $A \in \mathbb{S}^n$, $\mathbf{b} \in \mathbb{R}^n$, $c \in \mathbb{R}$, and $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be the quadratic function

$$f(\mathbf{x}) = \mathbf{x}^\top A \mathbf{x} + \mathbf{b}^\top \mathbf{x} + c = \sum_{i=1}^n \sum_{j=1}^n a_{ij} x_i x_j + \sum_{i=1}^n b_i x_i + c$$

From our work in the last lecture, and from the linear combination rule, the derivative of f is

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{h} \mapsto 2\mathbf{x}^\top A \mathbf{h} + \mathbf{b}^\top \mathbf{h},$$

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{h} \mapsto 2\mathbf{x}^\top A\mathbf{h} + \mathbf{b}^\top \mathbf{h},$$

Note that this is consistent with the univariate case $(ax^2 + bx + c)' = 2ax + b$.

$$Df(\mathbf{x}) : \mathbb{R}^n \rightarrow \mathbb{R}, \quad \mathbf{h} \mapsto 2\mathbf{x}^\top A \mathbf{h} + \mathbf{b}^\top \mathbf{h},$$

The Jacobian and gradient at $\mathbf{x}$ are

$$\begin{aligned} Jf(\mathbf{x}) &= 2\mathbf{x}^\top A^\top + \mathbf{b}^\top, \\ \nabla f(\mathbf{x}) &= 2A\mathbf{x} + \mathbf{b} \end{aligned}$$

The Jacobian here is the transpose of the gradient. This will be true in general only for scalar-valued functions.

What is the gradient of $\mathbf{x}^\top A \mathbf{x}$ when $A \in \mathbb{R}^{n \times n}$ is **not** symmetric?

Every matrix can be written as the sum of a symmetric matrix and a skew symmetric matrix:

$$A = \underbrace{\frac{A + A^\top}{2}}_{\text{symmetric}} + \underbrace{\frac{A - A^\top}{2}}_{\text{skew-symmetric}} = B + C$$

There is something special about skew-symmetric matrices, which will let us simplify the gradient calculation.

$$A = \underbrace{\frac{A + A^\top}{2}}_{\text{symmetric}} + \underbrace{\frac{A - A^\top}{2}}_{\text{skew-symmetric}} = B + C$$

Remember, a skew-symmetric matrix satisfies $C^\top = -C$.

Note that $\mathbf{x}^\top C \mathbf{x}$ is a scalar - a 1×1 matrix. So it must be equal to its transpose:

$$(\mathbf{x}^\top C \mathbf{x})^\top = \mathbf{x}^\top C \mathbf{x}$$

But we also know from the properties of the transpose that

$$(\mathbf{x}^\top C \mathbf{x})^\top = \mathbf{x}^\top C^\top \mathbf{x}$$

Since C is skew-symmetric, we have $C^\top = -C$. Putting this all together, we get

$$\mathbf{x}^\top C \mathbf{x} = (\mathbf{x}^\top C \mathbf{x})^\top = \mathbf{x}^\top C^\top \mathbf{x} = -\mathbf{x}^\top C \mathbf{x}$$

The only way this can be true is if $\mathbf{x}^\top C \mathbf{x} = 0$ for all $\mathbf{x} \in \mathbb{R}^n$.

So only the symmetric part of A contributes to $\mathbf{x}^\top A \mathbf{x}$.

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top B \mathbf{x} + \mathbf{x}^\top C \mathbf{x} = \mathbf{x}^\top B \mathbf{x} + 0 = \mathbf{x}^\top \left(\frac{A + A^\top}{2} \right) \mathbf{x}.$$

In summary, we always have

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top \left(\frac{A + A^\top}{2} \right) \mathbf{x} \tag{1}$$

for all $\mathbf{x} \in \mathbb{R}^n$.

So if A is not symmetric, we just need to rewrite $\mathbf{x}^\top A \mathbf{x}$ in symmetric form $\mathbf{x}^\top (A + A^\top) \mathbf{x} / 2$

using Equation 1 and we immediately get $Df(\mathbf{u}) : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{x} \mapsto (A + A^\top) \mathbf{x}$ and that

$$\nabla f(\mathbf{x}) = (A + A^\top) \mathbf{x}$$

Note that if A is symmetric, we get back $\nabla f(\mathbf{x}) = 2A\mathbf{x}$.

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{x} \mapsto \|\mathbf{x}\|^2$.

Note that $f(\mathbf{x}) = \mathbf{x}^\top \mathbf{x}$ and so this is just the previous example with $A = I, \mathbf{b} = \mathbf{0}, c = 0$. In particular

$$\nabla \|\mathbf{x}\|^2 = 2\mathbf{x}.$$

What about the norm function $\|\cdot\| : \mathbb{R}^n \rightarrow \mathbb{R}, \mathbf{x} \mapsto \|\mathbf{x}\|$?

We just use the chain rule with the square root function $g : \mathbb{R} \rightarrow \mathbb{R}, x \mapsto \sqrt{x}$ and $\|\cdot\| = g \circ f$. You will be asked to do this in Homework 3.

Simple example applications of the Chain Rule

Let $f : \mathbb{R}^m \rightarrow \mathbb{R}$ be the log-sum-exp function defined by

$$f(\mathbf{y}) = \log(e^{y_1} + \dots + e^{y_m})$$

We would like to optimize a function of the form $g(\mathbf{x}) = f(A\mathbf{x} + \mathbf{b})$.

Let $\mathbf{a}_1, \dots, \mathbf{a}_m \in \mathbb{R}^n$ and $b_1, \dots, b_m \in \mathbb{R}$. We set

$$A = \begin{bmatrix} \mathbf{a}_1^\top \\ \vdots \\ \mathbf{a}_m^\top \end{bmatrix} \in \mathbb{R}^{m \times n}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ \vdots \\ b_m \end{bmatrix} \in \mathbb{R}^m$$

Then the function $g : \mathbb{R}^n \rightarrow \mathbb{R}$,

$$g(\mathbf{x}) = f(A\mathbf{x} + \mathbf{b}) = \log \left(\sum_{i=1}^m \exp(\mathbf{a}_i^\top \mathbf{x} + b_i) \right)$$

The gradient of f is simple enough that the dumb formula can be applied directly (very rare in this course).

We just use the chain rule to find each of the partial derivatives, remembering that $\frac{\partial}{\partial z} \log(z) = 1/z$:

$$\frac{\partial f}{\partial y_j} = \frac{1}{\sum_{i=1}^m e^{y_i}} \cdot e^{y_j}$$

Then the gradient is just the vector of partial derivatives:

$$\nabla f(\mathbf{y}) = \frac{1}{\sum_{i=1}^m e^{y_i}} \begin{bmatrix} e^{y_1} \\ \vdots \\ e^{y_m} \end{bmatrix}$$

By the chain rule, where $g(\mathbf{x}) = f(A\mathbf{x} + \mathbf{b})$:

$$\nabla g(\mathbf{x}) = A^T \cdot \nabla f(A\mathbf{x} + \mathbf{b})$$

Now we substitute what we know about ∇f from earlier:

$$\nabla g(\mathbf{x}) = A^T \cdot \frac{1}{\sum_{i=1}^m e^{(A\mathbf{x}+\mathbf{b})i}} \begin{bmatrix} e^{(A\mathbf{x}+\mathbf{b})1} \\ \vdots \\ e^{(A\mathbf{x}+\mathbf{b})m} \end{bmatrix}$$

Optimizing such functions, that is,

$$\begin{aligned} & \text{minimize } g_0(\mathbf{x}) \\ & \text{subject to } g_i(\mathbf{x}) = 0, \quad i = 1, \dots, p \\ & \quad \quad \quad g_i(\mathbf{x}) \leq 0, \quad i = p + 1, \dots, r \end{aligned}$$

where $g_0, g_1, \dots, g_r$ are all functions of such form, is called **geometric programming** and is very important in circuit and chip designs.

The log-sum-exp function f arises as a smooth approximation of the nondifferentiable max function as

$$\max(x_1, \dots, x_n) \leq \text{lse}(x_1, \dots, x_n) \leq \max(x_1, \dots, x_n) + \log n.$$

You might wish to prove this as an exercise.

Review on symmetric, positive semidefinite, positive definite matrices

Recall that a matrix $A \in \mathbb{R}^{n \times n}$ is **positive semidefinite** if $\mathbf{x}^\top A \mathbf{x} \geq 0$ for all $\mathbf{x} \in \mathbb{R}^n$ and is **positive definite** if $\mathbf{x}^\top A \mathbf{x} > 0$ for all $0 \neq \mathbf{x} \in \mathbb{R}^n$.

The following notations are commonly used in optimization literature.

For the **sets** of symmetric etc matrices:

- Symmetric matrices: $\mathbb{S}^n = \{X \in \mathbb{R}^{n \times n} : X^\top = X\}$
- Symmetric positive semidefinite: $\mathbb{S}_+^n = \{X \in \mathbb{S}^n : X \text{ is positive semidefinite} \}$
- Symmetric positive definite: $\mathbb{S}_{++}^n = \{X \in \mathbb{S}^n : X \text{ is positive definite} \}$

For individual **matrices** X :

- positive definite: $X \succ 0$,
- positive semidefinite: $X \succeq 0$,
- negative definite: $X \prec 0$,
- negative semidefinite $X \preceq 0$.

So $X \in \mathbb{S}_+^n$ iff $X \succeq 0$ or $0 \preceq X$; $X \in \mathbb{S}_{++}^n$ iff $X \succ 0$ or $0 \prec X$.

- $\text{GL}(n) = \{A \in \mathbb{R}^{n \times n} : \det(A) \neq 0\}$ is the set of all $n \times n$ invertible matrices (the General Linear group).

The positive-definiteness of a matrix only depends on the symmetric part of the matrix.

We can add an arbitrary skew-symmetric matrix to a positive definite matrix and it will remain positive definite.

Why?

Let $A \in \mathbb{R}^{n \times n}$. Recall that for a skew-symmetric matrix C , i.e., $C^\top = -C$, we always have $\mathbf{x}^\top C \mathbf{x} = 0$ (why?), so

$$\mathbf{x}^\top A \mathbf{x} = \mathbf{x}^\top (A + C) \mathbf{x}$$

This means that knowing that a matrix is positive definite does not tell us anything about whether it is symmetric.

Some facts about symmetric positive (semi)definite matrices

- We will often want to be working with positive definite matrices
 - We'll be able to show that critical points of functions are true local minima if the Hessian is positive definite
- For symmetric matrices, because the eigenvalues are real, we'll be able to define positive definiteness in terms of eigenvalues being positive or negative – this is an easy condition to check
- We'll want to show that if we make small perturbations to a positive definite matrix, we still get a positive definite matrix

Eigenvalues of positive (semi)definite matrices

- If $X \in \mathbb{S}^n$, then the eigenvalues of X are real.
 - This lets us define positive definiteness in terms of eigenvalues being positive or negative
- If $X \in \mathbb{S}^n$, then
 - $X \in \mathbb{S}_+^n$ iff all eigenvalues are nonnegative;
 - $X \in \mathbb{S}_{++}^n$ iff all eigenvalues are positive;
- If $X \in \mathbb{S}^n$ is invertible, then $X^{-1} \in \mathbb{S}^n$.

A reminder: the k th leading principal minor of $X \in \mathbb{R}^{n \times n}$ is the determinant of the $k \times k$ matrix that is the upper left corner of X .

Sylvester's criterion: $X \in \mathbb{S}^n$ is positive definite iff all leading principal minors are positive.

(We could make a similar statement for positive semidefinite matrices, but it involves all principal minors, not just leading principal minors, so it's more complicated.)

- $\mathbb{S}_{++}^n$ is an open set in $\mathbb{S}^n$. (We will prove this today.)
 - This lets us show that if we make small perturbations to a positive definite matrix, we still get a positive definite matrix
- The closure of $\mathbb{S}_{++}^n$ is $\mathbb{S}_+^n$.

If we want to always be working with a positive definite (or semidefinite) matrices, we can do constrained optimization – but this can be a pain.

Another approach is to work with the space of all $n \times n$ matrices, and use the fact that these matrices times their transpose give positive semidefinite matrices.

- If $X \in \mathbb{S}^n$, then
 - $X \in \mathbb{S}_+^n$ iff $X = RR^\top$ for some $R \in \mathbb{R}^{n \times n}$;
 - $X \in \mathbb{S}_{++}^n$ if $X = RR^\top$ for some $R \in \text{GL}(n)$ (i.e. R is invertible).

Proof that $\mathbb{S}_{++}^n$ is open in $\mathbb{S}^n$

The set $\mathcal{S}_{++}^n$ is open in $\mathcal{S}^n$ since it can be written as the finite intersection of open sets:

$$\mathcal{S}_{++}^n = \{X \in \mathcal{S}^n : X \text{ is positive definite} \}$$

First, we will use Sylvester's criterion: a matrix is positive definite if and only if all its leading principal minors are positive.

This means that it must be in the set of matrices where the first leading principal minor is positive, and also in the set of matrices where the second leading principal minor is positive, and so on.

This is equivalent to the intersection of the sets of matrices where the determinants of the leading principal minors are positive:

$$\mathbb{S}_{++}^n = \bigcap_{k=1}^n \{X \in \mathbb{S}^n : \det(X_k) > 0\}$$

Now we will rewrite the condition that the k th leading principal minor is positive in terms of a *function of* X that computes that principal minor.

Call the function $m_k(X)$:

$m_k(X) := \det(X_k)$ is the k th principal minor of X .

$$\mathbb{S}_{++}^n = \bigcap_{k=1}^n \{X \in \mathbb{S}^n : \det(X_k) > 0\} = \bigcap_{k=1}^n \{X \in \mathbb{S}^n : m_k(X) > 0\}$$

For our next step, we would like to prove that the set of matrices where $m_k(X) > 0$ is open.

To do this, we will use the fact that $m_k(X)$ is a continuous function of X .

As we showed in an earlier lecture, the preimage of an open set under a continuous function is open.

OK. Saying that $m_k(X) > 0$ for a given X is equivalent to saying that $m_k(X)$ is in the open interval $(0, \infty)$.

The set of all matrices where $m_k(X) > 0$ is the preimage of the open set $(0, \infty)$ under the continuous function m_k .

The set of all matrices where $m_k(X) > 0$ is the preimage of the open set $(0, \infty)$ under the continuous function m_k .

$$\{X \in \mathbb{S}^n : m_k(X) > 0\} = m_k^{-1}((0, \infty)) \implies \text{the set is open}$$

So we've shown that the set of all matrices where $m_k(X) > 0$ is open.

Now we were working with the intersection of all such sets from $k = 1$ to n . We know that the intersection of finitely many open sets is open.

$$\bigcap_{k=1}^n \{X \in \mathbb{S}^n : m_k(X) > 0\} = \bigcap_{k=1}^n m_k^{-1}((0, \infty)) \implies \text{is open}$$

Similarly, the set of all positive semidefinite matrices is a closed subset of $\mathbb{S}^n$.

Whew! We've shown that the positive definite matrices form an open set in $\mathbb{S}^n$.

That means that if we have a positive definite symmetric matrix, and we make a small perturbation to it, we still get a positive definite symmetric matrix.

If the set had been closed, then this would not be true. If we happened to be at a boundary point, and made a small perturbation, we might leave the set and get a matrix that is not positive definite.

We'll be able to use this to optimize near critical points of functions, if we can show that the Hessian is positive definite at those points.

Showing that we can write positive definite matrices as a product of a matrix and its transpose

It follows from the spectral theorem for symmetric matrices that X has an eigendecomposition $X = Q\Lambda Q^\top$, where

$$Q^\top Q = I, \quad Q = [q_1, \dots, q_n], \quad \Lambda = \text{diag}(\lambda_1, \dots, \lambda_n), \quad Aq_i = \lambda_i q_i$$

Since $\lambda_i \geq 0$ for all i , we may take square roots and get

$$X = \underbrace{Q \text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n})}_{=R} \cdot \underbrace{\text{diag}(\sqrt{\lambda_1}, \dots, \sqrt{\lambda_n}) Q^\top}_{=R^\top}.$$

Hence $X = RR^\top$ for some $R \in \mathbb{R}^{n \times n}$.

Conversely, if $X = RR^\top$ then $X^\top = (RR^\top)^\top = (R^\top)^\top R^\top = RR^\top = X$.

Now for any $\mathbf{a} \in \mathbb{R}^n$,

$$\mathbf{a}^\top X \mathbf{a} = \mathbf{a}^\top (R^\top R) \mathbf{a} = (R\mathbf{a})^\top (R\mathbf{a}) = \|R\mathbf{a}\|^2 \geq 0$$

This is the same as saying that X is positive semidefinite.

The proof for positive definite matrices only requires a slight modification.

For nonzero $\mathbf{a} \in \mathbb{R}^n$, we have $R\mathbf{a} \neq 0$, so $\|R\mathbf{a}\| \neq 0$, and consequently, $\mathbf{a}^\top X \mathbf{a} > 0$.

Positive definiteness and positive semidefiniteness are commonly used to define constraints in optimization problem. For example, the follow type of constraints we see in Homework 3 is very common in control theory:

$$X^{\top}AX + B^{\top}X + X^{\top}B + C \succ 0.$$

Here X is an $m \times n$ matrix variable and $A \in \mathbb{S}^m, B \in \mathbb{R}^{m \times n}, C \in \mathbb{S}^n$ are constant matrices.

Derivatives of matrix-variate and matrix-valued functions

Let's start with a warm-up exercise.

Exercise

Extend **Exm-7p1** to matrix-variate functions, i.e., where the variable X is an $m \times n$ matrix. In particular, find the gradient of

$$\text{tr}(X^\top AX) + \text{tr}(B^\top X) + c$$

for $A \in \mathbb{R}^{m \times m}$, $B \in \mathbb{R}^{m \times n}$, $c \in \mathbb{R}$. Deduce that

$$\nabla \|X\|^2 = 2X$$

In this example, there is no dumb formula one can use. The trick is to look for the linear terms.

Let $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}, X \mapsto X^2$. We determine its derivative from definition:

$$\begin{aligned}\lim_{H \rightarrow 0} \frac{\|f(X+H) - f(X) - [Df(X)]H\|}{\|H\|} &= \lim_{H \rightarrow 0} \frac{\|(X+H)^2 - X^2 - [Df(X)]H\|}{\|H\|} \\ &= \lim_{H \rightarrow 0} \frac{\|X^2 + XH + HX + H^2 - X^2 - [Df(X)](H)\|}{\|H\|} \\ &= \lim_{H \rightarrow 0} \frac{\|XH + HX + H^2 - [Df(X)](H)\|}{\|H\|}\end{aligned}$$

$$= \lim_{H \rightarrow 0} \frac{\|XH + HX + H^2 - [Df(X)](H)\|}{\|H\|}$$

If we set $[Df(X)]H = XH + HX$, we get

$$\begin{aligned} &= \lim_{H \rightarrow 0} \frac{\|H^2\|}{\|H\|} \\ &\leq \lim_{H \rightarrow 0} \frac{\|H\|^2}{\|H\|} = \lim_{H \rightarrow 0} \|H\| = 0. \end{aligned}$$

Note that $\|H^2\| \leq \|H\|^2$ because we may assume that the norm is submultiplicative (since all norms are equivalent on $\mathbb{R}^{n \times n}$). So $Df(X)$ is the linear map

$$Df(X) : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}, \quad H \mapsto XH + HX$$

Exercise

Let $A, B \in \mathbb{R}^{n \times n}$. To practice the chain rule, find $Df(X)$ for the function $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}, X \mapsto (AX + B)^2$.

Exercise

Find $Df(X)$ for the function $f : \mathbb{R}^{n \times n} \rightarrow \mathbb{R}^{n \times n}, X \mapsto X^3$. As a challenge, do X^p for any integer $p > 1$ - this will come in handy for the derivative of $\exp(X)$ in Homework 3.

If you use the dumb formula for gradient below, you see that you get stuck very quickly. Again the trick is use the definition directly and look for the linear terms.

What is the derivative of the trace of the inverse of a positive definite matrix?

Let $f : \mathbb{S}_{++}^n \rightarrow \mathbb{R}, X \mapsto \text{tr}(X^{-1})$. We want $[Df(X)](H)$ so that the following quotient goes to zero as $H \rightarrow 0$:

$$\frac{|\text{tr}[(X+H)^{-1}] - \text{tr}(X^{-1}) - [Df(X)](H)|}{\|H\|}.$$

Well, this looks like a mess!

How will we deal with the inverse?

We remember that $(I - A)^{-1} = I + A + A^2 + A^3 + \dots$. This is helpful because it gets rid of the inverse, *and* we may be able to eliminate the higher order terms.

So let's try to get $X + H$ in the form $I + A$.

So let's try to get $X + H$ in the form $I + A$.

We can do this by pulling out X from the expression:

$$X + H = X (I + (X^{-1}H))$$

Now we use the formula for the inverse of a product $AB^{-1} = B^{-1}A^{-1}$ to get X over to the right side:

$$X (I + X^{-1}H)^{-1} = (I + X^{-1}H)^{-1} X^{-1}$$

$$X(I + X^{-1}H)^{-1} = (I + X^{-1}H)^{-1} X^{-1}$$

Now we expand the inverse:

$$(I + X^{-1}H)^{-1} X^{-1} = X^{-1} - X^{-1}HX^{-1} + (X^{-1}H)^2 X^{-1} - (X^{-1}H)^3 X^{-1} + (X^{-1}H)^4 X^{-1} - \dots$$

Now we take the trace (since that's what we started with):

$$\text{tr}[(X + H)^{-1}] = \text{tr}\left[X^{-1} - X^{-1}HX^{-1} + (X^{-1}H)^2X^{-1} - (X^{-1}H)^3X^{-1} + (X^{-1}H)^4X^{-1} - \dots\right]$$

Since the trace is linear, we can pull it through the sum:

$$= \text{tr}\left(X^{-1} - X^{-1}(HX^{-1}) + \text{tr}\left[(X^{-1}H)^2 - (X^{-1}H)^3 + (X^{-1}H)^4 - \dots\right]X^{-1}\right]$$

Now we pull out a factor of $(X^{-1}H)^2$ from each term in the rightmost sum:

$$= \text{tr}(X^{-1}) - \text{tr}(X^{-1}HX^{-1}) + \text{tr}\left[(X^{-1}H)^2\left(I - X^{-1}H + (X^{-1}H)^2 - \dots\right)X^{-1}\right]$$

$$= \text{tr} (X^{-1}) - \text{tr} (X^{-1} H X^{-1}) + \text{tr} \left[(X^{-1} H)^2 (I + X^{-1} H)^{-1} X^{-1} \right]$$

$$= \operatorname{tr}(X^{-1}) - \operatorname{tr}(X^{-1}HX^{-1}) + \operatorname{tr}\left[(X^{-1}H)^2(I + X^{-1}H)^{-1}X^{-1}\right]$$

So now our expression for the limit is:

$$\begin{aligned} &= \frac{|\operatorname{tr}[(X + H)^{-1}] - \operatorname{tr}(X^{-1}) - [Df(X)](H)|}{\|H\|} \\ &= \frac{|\operatorname{tr}(X^{-1}) - \operatorname{tr}(X^{-1}HX^{-1}) + \operatorname{tr}\left[(X^{-1}H)^2(I + X^{-1}H)^{-1}X^{-1}\right] - \operatorname{tr}(X^{-1}) - [Df(X)](H)|}{\|H\|} \end{aligned}$$

$$= \frac{\left| \operatorname{tr}(X^{-1}) - \operatorname{tr}(X^{-1}HX^{-1}) + \operatorname{tr}\left[(X^{-1}H)^2(I + X^{-1}H)^{-1}X^{-1}\right] - \operatorname{tr}(X^{-1}) - [Df(X)](H) \right|}{\|H\|}$$

Therefore if we choose $[Df(X)](H) = -\operatorname{tr}(X^{-1}HX^{-1})$, then as $H \rightarrow 0$

$$\begin{aligned} \frac{\left| \operatorname{tr}(X + H)^{-1} - \operatorname{tr}(X^{-1}) - [Df(X)](H) \right|}{\|H\|} &= \frac{\left| \operatorname{tr}(X^{-1}H)^2(I + X^{-1}H)^{-1}X^{-1} \right|}{\|H\|} \\ &\leq \frac{\|X^{-1}\|^2 \|H\|^2 \|(I + X^{-1}H)^{-1}\| \|X^{-1}\|}{\|H\|} \\ &= \|H\|^2 \|X^{-1}\|^3 \|(I + X^{-1}H)^{-1}\| \rightarrow 0 \end{aligned}$$

So $Df(X)$ is the linear map

$$Df(X) : \mathbb{S}^n \rightarrow \mathbb{R}, \quad H \mapsto -\operatorname{tr}(X^{-1}HX^{-1})$$

If we want to find the gradient, use the fact that $\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA)$ to see that

$$[Df(X)](H) = -\text{tr}(X^{-1}HX^{-1}) = \text{tr}\left[(-X^{-2})^{\top} H\right]$$

But this is just the Frobenius inner product:

$$\langle A, B \rangle_F = \text{tr}(A^T B) = \sum_{i=1}^m \sum_{j=1}^n A_{ij} B_{ij}$$

and thus

$$[Df(X)](H) = -\text{tr}(X^{-1}HX^{-1}) = \text{tr}\left[(-X^{-2})^{\top} H\right] = \langle -X^{-2}, H \rangle$$

Remember that in the special case $\mathbb{V} = \mathbb{R}$, the gradient of a function $f : \Omega \rightarrow \mathbb{R}$ at $u \in \Omega \subseteq \mathbb{U}$ is defined to be $\nabla f(u) \in \mathbb{U}$ such that

$$[Df(u)](h) = \langle \nabla f(u), h \rangle$$

where $\langle \cdot, \cdot \rangle : \mathbb{U} \times \mathbb{U} \rightarrow \mathbb{R}$ is the inner product on $\mathbb{U}$.

So we can finally identify the gradient (of the trace of the inverse of a matrix)!

$$\nabla f(X) = -X^{-2}$$

The reason we can make such a definition is because any linear map $T : \mathcal{U} \rightarrow \mathbb{R}$, i.e., codomain is the field of scalars, must take the form

$$T(h) = \langle t, h \rangle$$

for some unique $t \in \mathcal{U}$. This is called Riesz representation theorem and is an easy result to prove.

Proof of Riesz representation theorem

Take an orthonormal basis $u_1, \dots, u_n$ of $\mathbb{U}$. Then any $h \in \mathbb{U}$ can be written $h = \langle u_1, h \rangle u_1 + \dots + \langle u_n, h \rangle u_n$ and so

$$\begin{aligned} T(h) &= T(\langle u_1, h \rangle u_1 + \dots + \langle u_n, h \rangle u_n) \\ &= \langle u_1, h \rangle T(u_1) + \dots + \langle u_n, h \rangle T(u_n) && \text{(linearity of } T) \\ &= \langle \underbrace{T(u_1) u_1 + \dots + T(u_n) u_n}_t, h \rangle && \text{(bilinearity of inner product)} \end{aligned}$$

We just pick $t = T(u_1)u_1 + \cdots + T(u_n)u_n$.

Getting gradients from the derivative, more generally

- The gradient can be defined for any linear map whose codomain is the field of scalars, i.e. $T : \mathbb{U} \rightarrow \mathbb{R}$
- The gradient's codomain will match the function domain: If $f : \Omega \rightarrow \mathbb{R}$ maps from some space Ω to real numbers, and $\Omega \subseteq \mathbb{U}$ where $\mathbb{U}$ is a vector space, then its gradient ∇f maps from Ω back to the same space: $\nabla f : \Omega \rightarrow \mathbb{U}$

(sketch)

- Different possible spaces: The gradient's form depends on what space $\mathbb{U}$ is:
- When $\mathbb{U} = \mathbb{R}^n$: gradient is a vector
- When $\mathbb{U} = \mathbb{R}^{m \times n}$: gradient is a matrix
- When $\mathbb{U} = \mathbb{S}^n$: gradient is a symmetric matrix

In every case, the gradient and derivative are connected through the inner product:
 $Df(u) = \langle \nabla f(u), h \rangle$. We just need to use the correct inner product for the space $\mathbb{U}$.

Table of gradients

SPACE U	GRADIENT FORM	INNER PRODUCT
$\mathbb{R}^n$ (vectors)	Vector	$x_1^\top x_2$
$\mathbb{R}^{m \times n}$ (matrices)	Matrix	$\text{tr}(A_1^\top A_2)$
$\mathbb{S}^n$ (symmetric)	Symmetric matrix	$\text{tr}(Y_1^\top Y_2)$
$\mathbb{R}^n \times \mathbb{S}^n$ (product)	(vector, sym. matrix)	$x_1^\top x_2 + \text{tr}(Y_1^\top Y_2)$

Complex space example: Consider the product space $\mathbb{U} = \mathbb{R}^n \times \mathbb{S}^n$ (vectors combined with symmetric matrices), and $\mathbb{V} = \mathbb{R}$

An element $(\mathbf{x}, Y) \in \mathbb{R}^n \times \mathbb{S}^n$ will have 1 column (from $\mathbf{x}$) plus n more columns (from Y), for a total of $n + 1$ columns.

It will also have n rows.

The inner product is the usual trace inner product:

$$\langle (\mathbf{x}_1, Y_1), (\mathbf{x}_2, Y_2) \rangle = \text{tr} \left((\mathbf{x}_1, Y_1)^\top (\mathbf{x}_2, Y_2) \right) = \mathbf{x}_1^\top \mathbf{x}_2 + \text{tr} (Y_1^\top Y_2)$$

Note that we get the usual norm from this inner product:

$$\langle (\mathbf{x}, Y), (\mathbf{x}, Y) \rangle = \|\mathbf{x}\|^2 + \|Y\|^2 = \|(\mathbf{x}, Y)\|^2$$

Example (Matrix-variate function)

Let $f : \mathbb{R}^n \times \mathbb{S}^n \rightarrow \mathbb{R}$, $(\mathbf{x}, Y) \mapsto \mathbf{x}^\top Y \mathbf{x}$. The derivative at $(\mathbf{x}, Y) \in \mathbb{R}^n \times \mathbb{S}^n$ is a linear map $Df(\mathbf{x}, Y) : \mathbb{R}^n \times \mathbb{S}^n \rightarrow \mathbb{R}$ and to find its expression, we simply look for the linear terms in

$$\begin{aligned} f(\mathbf{x} + \mathbf{h}, Y + H) &= (\mathbf{x} + \mathbf{h})^\top (Y + H)(\mathbf{x} + \mathbf{h}) \\ &= \mathbf{x}^\top Y \mathbf{x} + \mathbf{x}^\top Y \mathbf{h} + \mathbf{x}^\top H \mathbf{x} + \mathbf{x}^\top H \mathbf{h} + \mathbf{h}^\top Y \mathbf{x} + \mathbf{h}^\top H \mathbf{x} + \mathbf{h}^\top Y \mathbf{h} + \mathbf{h}^\top H \mathbf{h} \\ &= f(\mathbf{x}, Y) + 2\mathbf{x}^\top Y \mathbf{h} + \mathbf{x}^\top H \mathbf{x} + 2\mathbf{x}^\top H \mathbf{h} + \mathbf{h}^\top Y \mathbf{h} + \mathbf{h}^\top H \mathbf{h} \end{aligned}$$

Therefore if we choose $[Df(\mathbf{x}, Y)](\mathbf{h}, H) = 2\mathbf{x}^\top Y \mathbf{h} + \mathbf{x}^\top H \mathbf{x}$, then

$$\begin{aligned} \frac{|f(\mathbf{x} + \mathbf{h}, Y + H) - f(\mathbf{x}, Y) - [Df(\mathbf{x}, Y)](\mathbf{h}, H)|}{\|(\mathbf{h}, H)\|} &= \frac{|2\mathbf{x}^\top H \mathbf{h} + \mathbf{h}^\top Y \mathbf{h} + \mathbf{h}^\top H \mathbf{h}|}{\sqrt{\|\mathbf{h}\|^2 + \|H\|^2}} \\ &\leq \frac{2\|\mathbf{x}\| \|H\| \|\mathbf{h}\| + \|\mathbf{h}\|^2 \|Y\| + \|\mathbf{h}\|^2 \|H\|}{\sqrt{\|\mathbf{h}\|_2^2 + \|H\|_F^2}} \end{aligned}$$

To see that the last expression goes to 0 as $(\mathbf{h}, H) \rightarrow (0, 0)$, write $a = \|\mathbf{h}\|, b = \|H\|$ to simplify notations, and note that as $(a, b) \rightarrow (0, 0)$,

$$\frac{ab}{\sqrt{a^2 + b^2}} \leq \frac{ab}{a} = b \rightarrow 0, \quad \frac{a^2}{\sqrt{a^2 + b^2}} \leq \frac{a^2}{a} = a \rightarrow 0, \quad \frac{a^2b}{\sqrt{a^2 + b^2}} \leq \frac{a^2b}{a} = ab \rightarrow 0$$

To find the gradient, we need to express $[Df(\mathbf{x}, Y)](\mathbf{h}, H)$ in terms of the inner product on $\mathbb{R}^n \times \mathbb{S}^n$,

$$[Df(\mathbf{x}, Y)](\mathbf{h}, H) = 2\mathbf{x}^\top Y \mathbf{h} + \mathbf{x}^\top H \mathbf{x} = (2Y\mathbf{x})^\top \mathbf{h} + \text{tr}(\mathbf{x}\mathbf{x}^\top H) = \langle (2Y\mathbf{x}, \mathbf{x}\mathbf{x}^\top), (\mathbf{h}, H) \rangle$$

giving us

$$\nabla f(\mathbf{x}, Y) = (2Y\mathbf{x}, \mathbf{x}\mathbf{x}^\top) \in \mathbb{R}^n \times \mathbb{S}^n$$

In case you're wondering why $\mathbf{x}^\top H \mathbf{x} = \text{tr}(\mathbf{x}\mathbf{x}^\top H)$: Note that $\mathbf{x}^\top H \mathbf{x}$ is a 1×1 matrix so $\mathbf{x}^\top H \mathbf{x} = \text{tr}(\mathbf{x}^\top H \mathbf{x})$. The desired equality then follows from the cyclic invariance of trace, i.e.,

$$\text{tr}(ABC) = \text{tr}(CAB) = \text{tr}(BCA) \quad \text{for any } A \in \mathbb{R}^{m \times n}, B \in \mathbb{R}^{n \times p}, C \in \mathbb{R}^{p \times m}$$